

Componente Curricular: Física - Dinâmica

Professor(a): *Isadora C. Pádua*

Série: 2ª

Data: *11/10*

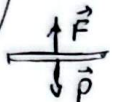
Conteúdo: *Potência e rendimento - Ex. Livros - Resolução*

10. a) $v_{ck} \rightarrow F_R = 0$

b) $\begin{array}{c} \uparrow \vec{F} \\ \downarrow \vec{p} \end{array}$ $F = P$ $G_F = F \cdot h$
 $F = 50N$ $G_F = 50 \cdot 10 \rightarrow G_F = 500J$

c) $Pot_u = \frac{G_u}{\Delta t} \Rightarrow 125 = \frac{500}{\Delta t}$
 $\Delta t = \frac{500}{125} \Rightarrow \Delta t = 4s$

11. * Diagrama $h \times t \rightarrow ret \rightarrow MU (v_{ck})$



$F = P \Rightarrow F = 1200N$

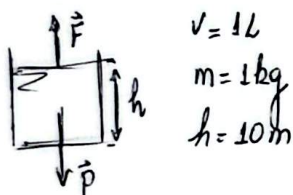
$Pot = \frac{G}{\Delta t} \Rightarrow Pot = \frac{F \cdot h}{\Delta t} \Rightarrow Pot = \frac{1200 \cdot 6}{20} \Rightarrow Pot = 360W$

14. * Energia liberada = $12MJ = 12 \cdot 10^6 J$ ($1MJ = 10^6 J$)

$E = G \Rightarrow G = 12 \cdot 10^6 J$

* $Pot = \frac{G}{\Delta t} \Rightarrow 2 \cdot 10^3 = \frac{12 \cdot 10^6}{\Delta t} \Rightarrow \Delta t = \frac{12 \cdot 10^6}{2 \cdot 10^3} = 6 \cdot 10^3 s = 3600s$
 $\Delta t = 1h$

16. $Pot_t = 4HP \Rightarrow Pot = 3000W$



$v = 1L$
 $m = 1kg$
 $h = 10m$

* $v_{ck} \Rightarrow F_R = 0$ ($F = P$)

* $1p = 15L$

* $1kg - 1L$
 $m - 15L$
 $m = 15kg$

* $Pot_u = \frac{F \cdot h}{\Delta t} \Rightarrow Pot_u = \frac{15 \cdot 10}{1} \Rightarrow Pot_u = 150W$

* $\eta = \frac{Pot_u}{Pot_t} \Rightarrow \eta = \frac{150}{3000} \Rightarrow \eta = 0,25$ (25%)

18. $Pot_t = 2500W$

$h = 15m$

$V = 9 \cdot 10^4 L$

$d = 1kg/L$

$m = 9 \cdot 10^4 kg$

$\Delta t = 5h = 18000s$

* Suposição: $v_{ck} (F_R = 0)$

$F = P \Rightarrow F = 9 \cdot 10^5 N$

* $Pot_u = \frac{G}{\Delta t}$

$Pot_u = \frac{F \cdot h}{\Delta t}$

$Pot_u = \frac{9 \cdot 10^5 \cdot 15}{18 \cdot 10^3}$

$Pot_u = 75 \cdot 10^2 W$

* $\eta = \frac{Pot_u}{Pot_t}$

$\eta = \frac{75 \cdot 10^2}{25 \cdot 10^3}$

$\eta = 0,5$ (50%)